Senior Design Test Plan

MedAssist

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Part I. Description of Overall Test Plan

Our project uses RAG (retrieval-augmented generation) architecture to support conversations related to mental health supported with information retrieved from DSM5 (The Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition). Specifically, the system ingests domain-specific documents, splits them into manageable chunks, enriches them with metadata, and embeds them into a vector for efficient retrieval. During user interactions, relevant document segments are retrieved and provided as context to a language model to reduce hallucination. As a result, our testing approach is organized around verifying each stage of this pipeline (data loading, chunking split, metadata enrichment, and embedding) and confirming that the query workflow reliably retrieves relevant content for response generation.

The overall test plan combines unit testing for component correctness with integration testing for end-to-end behavior. In addition to functional correctness, the plan also includes performance tests focused on ingestion scalability and query latency to make sure the system remains usable under realistic workloads and larger documents. Furthermore, all front-end features also undergo unit testing to ensure that all functions are accounted for and operable.

Part II. Test Case Descriptions

MA-01: Document Loading

- **Purpose:**
  Verify raw input documents (pdf, docx, and supported text formats) can be successfully loaded into the system for processing.
- **Description:**
  The system is provided with a set of valid document files specified in the input configuration. The document loader can read each file without error and contains non-empty textual content. This test ensures the ingestion pipeline can correctly access raw data before further processing such as chunking, metadata extraction, etc.
- **Inputs:**
  One or more valid document files located in the configured input directory.

- **Expected Outputs:**
  All input files are successfully loaded, each with unique identifier and extracted text content. No errors or warnings are generated during loading.
- **Case Type:**
  Normal
- **Test Type:**
  Blackbox
- **Test Category:**
  Functional
- **Test Level:**
  Unit

MA-02: Text Chunking

- **Purpose:**
  Verify that text is correctly split into chunks based on provided configurations.
- **Description:**
  The resulting chunks are inspected to confirm that they respect the specified configuration (chunk size, overlap, split strategy).
- **Inputs:**
  Text document, configuration such as chunk size, overlap, and split strategy parameters.
- **Expected Outputs:**
  Generated chunks follow the configuration limits, with no empty or excessively large chunks.
- **Case Type:**
  Boundary
- **Test Type:**
  Whitebox
- **Test Category:**
  Functional
- **Test Level:**
  Unit

MA-03: Summary Metadata Generation

- **Purpose:**
  Confirm that summary metadata is generated for each text chunk.
- **Description:**
  The ingestion pipeline is run with the summary extractor enabled, and the resulting nodes are examined for the presence of summary metadata.
- **Inputs:**
  One or more documents with sufficient textual content.
- **Expected Outputs:**
  Each generated node contains non-empty summary field describing the original text.
- **Case Type:**
  Normal
- **Test Type:**
  Whitebox
- **Test Category:**
  Functional
- **Test Level:**
  Unit

MA-04: Embedding Creation

- **Purpose:**
  Verify that vector embeddings are generated for all nodes.
- **Description:**
  After ingestion, the system generates embeddings for each node. The embeddings are inspected to ensure they exist and are consistent in dimensionality.
- **Inputs:**
  A small set of generated text nodes.
- **Expected Outputs:**
  All nodes contain valid embedding vectors with consistent dimensions.
- **Case Type:**
  Normal
- **Test Type:**
  Whitebox
- **Test Category:**

Functional

- **Test Level:**
  Unit

MA-05: Ingestion Cache Reuse

- **Purpose:**
  Ensure cached ingestion results are reused on repeated runs.
- **Description:**
  The ingestion pipeline is executed twice using identical inputs and an existing cache file to verify that re-computation is avoided (to save API costs).
- **Inputs:**
  Unchanged text documents with an existing ingestion cache.
- **Expected Outputs:**
  The second run uses cached results, completes faster, and produces identical nodes.
- **Case Type:**
  Normal
- **Test Type:**
  Whitebox
- **Test Category:**
  Performance
- **Test Level:**
  Integration

MA-06: Index Persistent and Reloading

- **Purpose:**
  Validate that the vector index can be saved and reloaded across sessions.
- **Description:**
  An index is built from ingested nodes, persisted to storage, and then reloaded after restarting the application.
- **Inputs:**
  Persisted index files in the configured storage directory.
- **Expected Outputs:**
  The index reloads successfully and is usable for subsequent queries without rebuilding.

- **Case Type:**
  Normal
- **Test Type:**
  Whitebox
- **Test Category:**
  Functional
- **Test Level:**
  Integration

MA-07: Basic Query Retrieval

- **Purpose:**
  Verify that the system retrieves relevant document content in response to user query.
- **Description:**
  A query with a known answer contained in the DSM5 document is submitted to the system. The retrieved nodes and generated responses are examined for relevance.
- **Inputs:**
  A user query whose answer appears explicitly in the source documents.
- **Expected Outputs:**
  The system retrieves relevant text chunks and produces a response related to the indexed material.
- **Case Type:**
  Normal
- **Test Type:**
  Blackbox
- **Test Category:**
  Functional
- **Test Level:**
  Integration

MA-08: Out-of-Scope Query Handling

- **Purpose:**
  Ensure the system handles queries not covered by the knowledge base without hallucinating.
- **Description:**

A user submits a question unrelated to the documents to test how the system responds when no relevant information is available.

- **Inputs:**
  A query outside the scope of DSM5.
- **Expected Outputs:**
  The system indicates insufficient information or requests clarification rather than generating fabricated details.
- **Case Type:**
  Abnormal
- **Test Type:**
  Blackbox
- **Test Category:**
  Functional
- **Test Level:**
  Integration

MA-09: Retrieval Ranking Accuracy

- **Purpose:**
  Validate that the most relevant document chunks are ranked highest.
- **Description:**
  A query with multiple relevant sections in the source documents is submitted. The top retrieved nodes are reviewed to confirm that the most relevant content appears first.
- **Inputs:**
  A query that matches several sections of the source documents.
- **Expected Outputs:**
  The top ranked retrieved nodes correspond to the most relevant sections of the documents.
- **Case Type:**
  Normal
- **Test Type:**
  Whitebox
- **Test Category:**
  Functional
- **Test Level:**
  Integration

MA-10: Query Response Time

- **Purpose:**
  Measure system responsiveness under typical usage conditions.
- **Description:**
  A set of representative queries is submitted sequentially while measuring response latency.
- **Inputs:**
  15-20 typical user queries.
- **Expected Outputs:**
  Average response times remain within acceptable limits for interactive use.
- **Case Type:**
  Normal
- **Test Type:**
  Blackbox
- **Test Category:**
  Performance
- **Test Level:**
  Integration

MA-11: Large Document Ingestion Performance

- **Purpose:**
  Evaluate ingestion scalability with large documents.
- **Description:**
  The ingestion pipeline is run on large documents (~1000 pages) to test memory usage and execution time.
- **Inputs:**
  The DSM5 document.
- **Expected Outputs:**
  Ingestion completes successfully without crashing, and resource usage remains within system limits.
- **Case Type:**
  Boundary
- **Test Type:**
  Blackbox

- **Test Category:**
  Performance
- **Test Level:**
  Integration

MA-12: End-to-end System Workflow

- **Purpose:**
  Validate the complete workflow from document ingestion to user response.
- **Description:**
  Starting from a clean environment, the system ingests documents, builds the index, and answers multiple user queries.
- **Inputs:**
  DSM5 document and a set of user queries.
- **Expected Outputs:**
  The system successfully completes the pipeline with correct responses and no runtime errors.
- **Case Type:**
  Normal
- **Test Type:**
  Blackbox
- **Test Category:**
  Functional
- **Test Level:**
  Integration

MA-13: Database Schema Validation

- **Purpose:**
  Verify that the database schema correctly supports required application data.
- **Description:**
  The database schema is reviewed and exercised by inserting and querying representative records to confirm that tables, fields, and relationships match the system design.
- **Inputs:**
  Sample user records, feedback entries, and journal data.
- **Expected Outputs:**

Data is stored and retrieved correctly.

- **Case Type:**
  Normal
- **Test Type:**
  Whitebox
- **Test Category:**
  Functional
- **Test Level:**
  Integration

MA-14: Backend-Database Integration

- **Purpose:**
  Ensure the backend correctly communicates with the database.
- **Description:**
  Backend services perform CRUD operations through the database interface to verity correct integration.
- **Inputs:**
  API calls that trigger database read/write operations.
- **Expected Outputs:**
  Backend operations succeed and database state updates accordingly.
- **Case Type:**
  Normal
- **Test Type:**
  Whitebox
- **Test Category:**
  Functional
- **Test Level:**
  Integration

MA-15: Frontend Accessibility and Usability

- **Purpose:**
  Ensure the frontend meets basic accessibility and usability standards.
- **Description:**
  The UI is tested for readable text, proper contrast, navigation, and clear interaction flow.

- **Inputs:**
  User interaction using mouse and keyboard across UI components.
- **Expected Outputs:**
  Interface remains usable without errors, accessibility issues are minimal.
- **Case Type:**
  Boundary
- **Test Type:**
  Blackbox
- **Test Category:**
  Functional
- **Test Level:**
  Integration

MA-16: End-to-end System Validation

- **Purpose:**
  Validate the system as a complete, integrated application.
- **Description:**
  The system is tested from initial document ingestion through user interaction, data storage, retrieval, and response generation.
- **Inputs:**
  DSM5 document, user queries, feedback entries, and journal data.
- **Expected Outputs:**
  All components function together correctly with consistent and reliable behavior.
- **Case Type:**
  Normal
- **Test Type:**
  Blackbox
- **Test Category:**
  Functional
- **Test Level:**
  Integration

MA-17: User Journal Functionality

- **Purpose:**
  Validate the front-end journaling functionality.

- **Description:**
  The journal page is tested for full functionality. A note/memory is submitted for a certain date. Additional reflection comments are added.
- **Inputs:**
  Notes/Memories (text), reflection comments (text).
- **Expected Outputs:**
  All journal components function correctly, and data is stored and retrieved properly.
- **Case Type:**
  Normal
- **Test Type:**
  Blackbox
- **Test Category:**
  Functional
- **Test Level:**
  Unit

MA-18: User Feedback Functionality

- **Purpose:**
  Ensure the feedback system works as intended.
- **Description:**
  The feedback page is tested by posting and responding to user submitted questions.
- **Inputs:**
  User questions (text), system administrator answers (text).
- **Expected Outputs:**
  Questions can be posted, send out notifications of submission, and are answerable from the system administrator view.
- **Case Type:**
  Normal
- **Test Type:**
  Blackbox
- **Test Category:**
  Functional
- **Test Level:**
  Unit

MA-19: Account Modification Functionality

- **Purpose:**
  Confirm account modifications are persistent and are reflected in the database.
- **Description:**
  User and administrator level accounts are modified (name change, password change, email, etc.). The associated database is checked to ensure all changes are stored as intended.
- **Inputs:**
  Account information to be changed.
- **Expected Outputs:**
  Database and front-end reflect all modifications.
- **Case Type:**
  Normal
- **Test Type:**
  Blackbox
- **Test Category:**
  Functional
- **Test Level:**
  Unit

MA-20: Reference Functionality

- **Purpose:**
  Verify that all references to third-party sources work.
- **Description:**
  Links to third-party mental health resources are tested to ensure that they are clickable on the front-end and direct to the proper web page(s).
- **Inputs:**
  Resource links.
- **Expected Outputs:**
  All links successfully direct to their web page.
- **Case Type:**
  Normal
- **Test Type:**
  Blackbox
- **Test Category:**

Functional

- **Test Level:**
  Unit

Part III. Test Case Matrix

	Case Type	Test Type	Test Category	Test Level
MA-01	Normal	Blackbox	Functional	Unit
MA-02	Boundary	Whitebox	Functional	Unit
MA-03	Normal	Whitebox	Functional	Unit
MA-04	Normal	Whitebox	Functional	Unit
MA-05	Normal	Whitebox	Performance	Integration
MA-06	Normal	Whitebox	Functional	Integration
MA-07	Normal	Blackbox	Functional	Integration
MA-08	Abnormal	Blackbox	Functional	Integration
MA-09	Normal	Whitebox	Functional	Integration
MA-10	Normal	Blackbox	Performance	Integration
MA-11	Boundary	Blackbox	Performance	Integration
MA-12	Normal	Blackbox	Functional	Integration
MA-13	Normal	Whitebox	Functional	Integration
MA-14	Normal	Whitebox	Functional	Integration
MA-15	Boundary	Blackbox	Functional	Integration
MA-16	Normal	Blackbox	Functional	Integration
MA-17	Normal	Blackbox	Functional	Unit
MA-18	Normal	Blackbox	Functional	Unit
MA-19	Normal	Blackbox	Functional	Unit
MA-20	Normal	Blackbox	Functional	Unit